

Solid High-Speed Synchronous Reluctance Rotor Enabled by Multi-Material Additive Manufacturing

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Abstract—Synchronous reluctance (SynR) motor technology is promising to realize rare-earth material-free electric machines. However, structural challenges limit operation speed and subsequently power density compared to machines with rare-earth permanent magnets. This paper proposes and investigates multi-material additive manufacturing (MMAM) as a key-enabler to realize power-dense SynR machines. It does so by guiding magnetic flux through a solid rotor component by selective placement of magnetic and non-magnetic materials to enable high-speed operation. To validate this concept, samples are manufactured using a MMAM process and experimentally characterized to assess the structural and magnetic properties that can be expected for the proposed rotors. The data is then used in a multi-physics modeling framework to explore the design space of new MMAM rotor concepts. The simulated results in this paper reveal that MMAM technology can enable a 4x increase in rotor speed, resulting in 400 % power density improvements. The MMAM rotors achieved tip speeds of approximately 300 m/s and rotational speeds over 55 kRPM at comparable efficiencies to conventional designs, despite the presence of existing MMAM geometry restrictions. This study ultimately demonstrates that MMAM technology has the potential to enhance SynR machine operation speed and power density, making it a valuable option for high-performance applications.

Index Terms—synchronous reluctance, additive manufacturing, multi-material additive manufacturing, 3D printed electric machines, optimization, power density

I. INTRODUCTION

With the advent of electrification, requirements towards improved energy use have demanded cleaner, more efficient electric machines. Synchronous reluctance (SynR) machines provide several benefits over their counterparts. This includes a lack of environmentally-harmful and expensive rare-earth permanent magnet (PM) material compared to PM machines, and higher torque density and efficiency compared to induction machines (IMs) [1]. However, SynR machines have limited speed and power density capabilities due to their complex rotor geometries [2], which result in reduced mechanical strength at high operation speeds [3] due to the presence of stress concentrations. Improving the strength of SynR rotors could yield power-dense machines, positioning SynR machines as promising rare-earth free alternatives to PM machines.

This paper proposes to enhance the mechanical strength of SynR machines by altering rotor flux barriers made of

air to instead be made of a strong, solid material. Additive manufacturing (AM) provides an avenue towards doing so. AM has been applied to the fabrication of electric machine components in different capacities, e.g.: [4], [5], resulting in improved machine performance metrics. Multi-material additive manufacturing (MMAM) is a new development within the AM field that utilizes multiple materials across the body of a single geometric structure [6]. This has a primary advantage that enables a single feature to contain different material properties across a continuous region. The ability to print and characterize MMAM samples for use in electric machines is the first step towards realizing MMAM's potential in the field of electric machines. Its application could provide a pathway to fully realize new SynR designs absent of the structural issues hindering conventional ones. This would allow SynR machines to operate at higher speeds and become more power-dense.

The key contributions of this study are i) the proposal of one of the first MMAM SynR rotors, ii) design space exploration for the proposed design, and iii) comparisons between the proposed design and a conventional baseline design. The paper initially reviews background information on SynR technology (Section II), MMAM technology (Section III), and material information and selections (Section IV). The paper then discusses the models and details the optimization framework used for three different rotor topologies in Section V. Results from the optimization are presented and selected designs are identified and analyzed in Section VI.

II. SYNCHRONOUS RELUCTANCE TECHNOLOGY

This section introduces and reviews SynR machine technology and identifies key metrics and performance parameters. Important characteristics to consider within SynR design include saliency ratios, which lead to torque production, and machine speed and power capabilities. Considering all of these characteristics is crucial in order to identify what performance metrics are important to achieve high-quality SynR rotor designs using MMAM.

A. Operating Principle

Speed and power density restrictions of conventional SynR machines result from their often-complex rotor geometries that utilize very small or thin geometric features. These features are required to maximize torque, but lead to high induced rotor stresses. Electromagnetic torque should be maximized

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to increase the power density and efficiency of SynR designs. Reluctance torque, T , of any electric machine is related to the number of pole pairs p , currents $i_{d,q}$, and axis inductances $L_{d,q}$, per (1).

$$T = 2p(L_d - L_q)i_d i_q \quad (1)$$

Reluctance torque is generated by employing a non-uniform rotor shape to create differences in the d- and q-axis inductances. To maximize reluctance torque, the difference between L_d and L_q must be maximized. In conventional SynR design, that leads to a complex and structurally-weak rotor. For instance, Fig. 1 shows two rotors with different saliency ratios (L_d/L_q). Figure 1a, has a lower saliency ratio than Fig. 1b, and if excited by the same current will produce less torque. However, Fig. 1a contains a stronger rotor geometry than Fig. 1b, and thus is able to operate at higher speeds. SynR machines are able to achieve comparable magnetic shear to other high-performance machines by having higher saliency ratios, despite having no rotor field source [7].

Existing publications have identified rotor topologies to maximize the saliency ratio [9], [10]. These publications find that maximum theoretical saliency ratios reach 20-25 and maximum practical saliency ratios reach 10-11, but with significant speed limitations. A process for finding optimal rotor designs for maximized torque production while maintaining structural rotor integrity is found in [11], resulting in a maximum practical saliency ratio of about 10.6. Designs producing higher torque have less structural integrity, resulting in lower speed ranges but higher torque capabilities. SynR machines that are designed for high speed typically have saliency ratios of up to 7-8 [11]. Both [12] and [13] find optimal flux barrier designs for high-speed applications, and [14] finds an ideal rotor topology to minimize torque ripple while minimizing effects on rotor structural integrity.

B. Machine Output Quantifications

Machine output power, P , is a key quantification because it directly relates to power density, which should be maximized. Additionally, the speed-power metric, $\omega\sqrt{P}$, indicates the maximum power that a machine can reasonably be designed for given a maximum operating speed [15]. Expressions for

power and speed-power can be derived as

$$P = T\omega = 2V_r\sigma_{F,\tan}\omega = 2\pi rL\sigma_{F,\tan}v_{\text{tip}} \quad (2)$$

$$\omega\sqrt{P} = \sqrt{2\pi\sigma_{F,\tan}v_{\text{tip}}^3\left(\frac{L}{r}\right)} \quad (3)$$

in terms of general machine design parameters of tip speed v_{tip} , rotor radius r , stack length L , and magnetic shear $\sigma_{F,\tan}$. Tip speed is an indication of how fast the edge of the rotor can travel, and is commonly used to quantify the structural and electrical performance of electric machines. Synchronous reluctance machines have maximum tip speeds of 130 - 170 m/s. This is lower than that of laminated, non-salient machines and IMs that can achieve tip speeds of 200 - 250 m/s [7]. The tip speed limitation in SynR machines restricts their usefulness because this limits rotational speed ω , which ultimately limits power and speed-power as seen in (2) and (3).

Given its tip speed restrictions, SynR machines are often used for slower-speed applications (4,000 - 6,000 RPM). Typical efficiencies for SynR machines are high at slower speeds, with > 95 % common in various applications [16]. The highest performance SynR machines occur at high slot/pole ratios, with the process described in [17] being the method of choice for the conventional rotor construction used in this study. Power (and thus power density) and the speed-power metric could be increased through innovation in the rotor design to increase structural integrity. This paper investigates how new MMAM techniques can be used to enhance SynR machine performance by mitigating sources of stress concentrations that restrict conventional designs.

III. MULTI-MATERIAL ADDITIVE MANUFACTURING TECHNOLOGY

This section reviews the applications of multi-material additive manufacturing (MMAM) to SynR technology and describes the printing process used in this paper. Several challenges of SynR rotor design can be addressed by using MMAM. This work focuses on the restrictions posed by conventional flux barrier design. However, the physical implementation of MMAM as a mean to eliminate flux-barriers poses its own design restrictions. This section explores these limitations and its impact on the proposed design in detail.

A. Application to Synchronous Reluctance Machines

Multi-material additive manufacturing (MMAM) is an AM method that utilizes multiple materials across the body of a single geometric feature [6]. Advancements in MMAM have seen an increase in recent years [4], and the field of electric machines contains many possibilities for its implementation. Existing literature proposes the replacement of air flux barriers in conventional SynR machines with a metallic insulator [4]. In this scenario, stainless steel with magnetic permeability properties comparable to air, is proposed as an ideal replacement.

The removal of the ribs in SynR rotors increases the power density and efficiency [16], [18], and improves the structural integrity of the rotor. The load that would act on the structural ribs is now spread out across the entirety of the magnetic

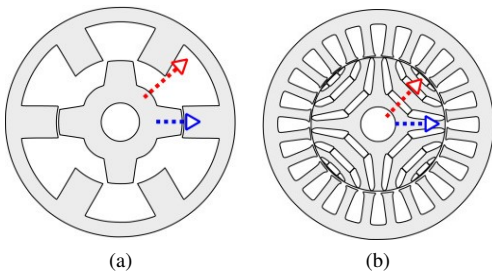


Fig. 1. Rotors with (a) smaller and (b) larger saliency ratios [8].

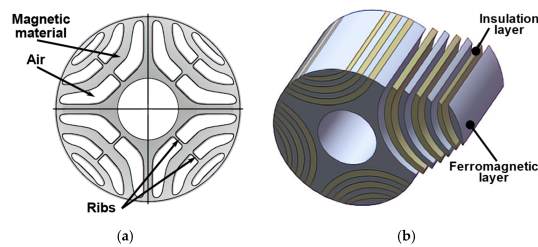


Fig. 2. Design of synchronous reluctance rotors [4].

flux barrier/flux carrier interface, drastically reducing the stress concentrations within the rotor. This concept is illustrated in both Fig. 2a, where the diagram of a conventional SynR rotor can be visualized and Fig. 2b, which diagrams a conceptual transversely-laminated rotor design with magnetic and insulation material layers. In this new design, maximum rotor speed and power density can be increased, therefore removing the two main limitations of SynR machines. MMAM technology can be used to fabricate this design since the process can form a metallic bond between the two materials and seamlessly integrate stainless steel flux barrier into the rotor design.

B. Physical Implementation and Restriction

The two main challenges of implementing MMAM in the field of electric machines are limited availability of commercial MMAM machines capable of handling at least two materials simultaneously and limited knowledge of bonding mechanisms of dissimilar material within MMAM context [19]. The AM process used in this work is laser powder bed fusion (LPBF). To create the solid structure via LPBF, a laser beam is used to deliver thermal energy to a powder bed in order to fully melt metal powder particles. The molten pool then rapidly cools to form a solid layer. This process is then repeated layer-by-layer to form a 3D object with exceptional geometrical accuracy [20]. An existing LPBF system is modified to incorporate MMAM capability. Figure 3 details the layout of the multi-material LPBF approach used in this work. The modified system has proven capable of creating metallic bonds between dissimilar materials [21]. This system is only capable of creating two dimensional interfaces in vertical or horizontal directions with respect to its build direction. This restriction, along with the machine's maximum build envelope, are the limiting factors in the proposed MMAM rotor designs. Future advancements will soon lift these restrictions and allow for more complex MMAM rotor designs [22].

IV. MATERIAL CHARACTERIZATION

This section details the decision making processes to select each MMAM material so that it may be printed and characterized. Both the magnetic material and flux barrier material are chosen to optimize the performance of the MMAM SynR machine. Each material should have independent properties optimal for their respective use. Once the materials are chosen, each is physically printed, using the aforementioned LPBF process, and characterized. These characterization results are

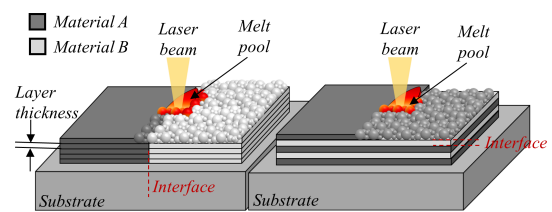


Fig. 3. Layout of the multi-material LPBF method used in this work for printing (a) vertical and (b) horizontal material interfaces.

used in FEA models documented later in the paper to investigate MMAM rotor concepts.

A. Material Selections

Magnetic material constitutes the bulk of the rotor and should have ideal properties for use in electric machines. High permeability, high saturation, low thermal resistivity, and high thermal conductivity are desired [23]. However, there is a trade-off since the two MMAM materials must be able to form a metallic bond. Fe-Si with 3.5 weight % Si (Fe-3.5Si) is chosen for this study because it has proven to be the most manufacturable and available magnetic material used for electric machine cores. Although increasing the Si content beyond 3.5 % can limit eddy current losses, studies have shown that Fe-Si with high silicon content is highly brittle and therefore difficult to manufacture [4], [23]. As such, Fe-3.5Si is the most practical both for implementation in an electric machine and as a printable material via LPBF [24]. The flux barrier material should have permeability similar to that of air. Furthermore, it should possess mechanical properties similar to that of Fe-3.5Si for adequate structural integrity. To ensure proper metallic bonding during the MMAM process, the flux barrier material should have similar crystal structure, coefficient of thermal expansion, Marangoni number, and manufacturability parameters to Fe-3.5Si [21]. The stainless steel 316L fits the above criteria and was therefore chosen as the flux barrier material.

B. Experimental Validation of MMAM Capability

The key manufacturing parameters in LPBF are laser power, laser speed, hatch spacing, and layer thickness. These parameters should be optimized to avoid the formation of defects such as porosity or cracks during LPBF. Defects can adversely affect mechanical and electromagnetic properties of the material [20]. Process parameter optimization for Fe-3.5Si and Fe-3.5Si-316L multi-material has been carried out experimentally. Test samples were successfully fabricated as shown in Figs. 4a and 4b. Detailed discussion of manufacturability and process parameter optimization is outside the scope of this paper and is the subject of another work in progress.

Since rotors are exposed to tension while operating, tensile tests are executed according to ASTM Standard EM/EM8 [25] to determine the mechanical properties of each material. Tensile tests were carried out on the single-material samples in order to verify their material properties. The results are shown

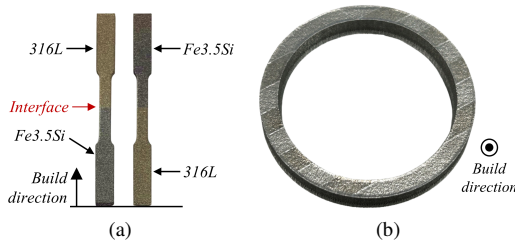


Fig. 4. (a) Multi-material Fe-3.5Si-316L tensile and (b) Fe-3.5Si ring test specimens

TABLE I
EXPERIMENTALLY MEASURED STRUCTURAL PROPERTIES

Mechanical Property	Fe-3.5Si	316L	Units
Young's Modulus	119	140	GPa
Yield Strength	495	460	MPa
Ultimate Strength	561	513	MPa

in Table I, where Poisson's ratio is assumed to be 0.33 for both materials.

The maximum speed of MMAM rotors is defined as the speed at which the weaker material reaches its yield strength. Therefore, it is important to verify that the MMAM process is able to create a multi-material bond that is stronger in tension than the material with the lower yield strength. Multi-material tensile specimens were fabricated (see Fig. 4a) and tested to determine if the material interface has an acceptable bond strength. Details of the test procedure can be found in [21]. Results show that necking and fracture does not occur at the interface of the two materials, proving the interfacial bond does not yield before the material with the lower yield strength. In other words, the highest stress throughout the multi-material rotors will occur in the 316L stainless steel material, not along the interfaces between the Fe-3.5Si and 316L. This provides confidence that the yield strength of 460 MPa can be used for the entire rotor during the optimization. These results indicate that current and future MMAM rotors are not compromised by Fe-3.5Si-316L bond strength.

Subsequently, single material ring samples were printed and tested to determine the BH curve of printed Fe-3.5Si. The samples were heat treated at 1000 C for 1 hour in vacuum conditions and tested according to ASTM Standards A773 [26] and A927 [27], which document the process to find the magnetic properties of materials under DC and AC conditions. The BH curve for DC is shown in Fig. 5, which is consistent with rotor conditions under synchronous operation. The printed Fe-3.5Si strongly resembles the popular electric steel M19 29 Gauge, which provides confidence it can be used in SynR rotors.

These results from both the tensile and ring tests are used in FEA models presented later, in Section VI, to explore the design space for the proposed MMAM rotor.

V. MULTI-PHYSICS OPTIMIZATION FRAMEWORK

This section formulates the optimization space and describes the multi-physics framework to investigate the potential of

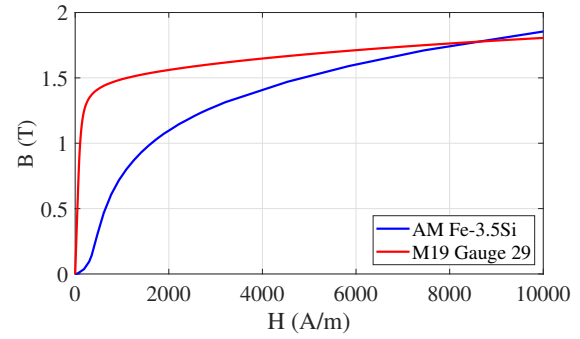


Fig. 5. BH curve for the Fe-3.5Si compared to a standard magnetic steel (M19).

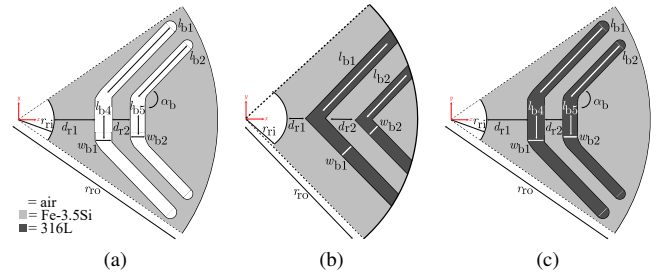


Fig. 6. Parameterized rotor cross-sections of: (a) baseline, (b) MMAM, and (c) vision rotor designs.

MMAM to improve SynR machine performance. Three rotor design topologies, shown in Fig. 6, are considered to represent a conventional flux barrier design ("baseline design"), a design with V-shaped flux barrier filled with stainless steel that can be built within the constraints of existing MMAM capabilities ("MMAM design"), and an ideal MMAM design that removes the constraints of today's MMAM printers ("vision design"). Comparisons between the baseline and MMAM designs will show the benefits and drawbacks of utilizing MMAM technology. Results from the vision rotor will show the potential of MMAM once the restrictions present in current MMAM technology are overcome.

A. Problem Formulation

A 36 slot stator design, with a double layer winding and a coil pitch of 8 was extracted from an existing motor and used without modification with each rotor design. The stator corresponds to a 145TC frame size and has an inner radius of 50.5 mm. Cross sections of the baseline, MMAM, and vision rotors are shown in Fig. 6. The general parameterization for each rotor is based on the design process outlined in [17] with the MMAM design altered to adapt to the MMAM/LPBF restrictions detailed in Section III-B. Information on rotor geometries is further given in Table II.

The design space for each rotor is explored within an optimization framework that utilizes a multi-objective evolutionary algorithm [28]. The objectives for this optimization are the practically-relevant metrics of power density, efficiency, and

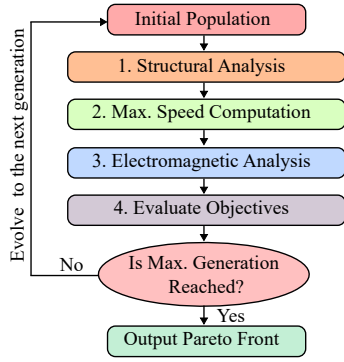


Fig. 7. Optimization flowchart

torque ripple. The free variables for this optimization are the rotor geometric parameters shown in Fig. 6.

The optimization varies the free variables listed in Table II within their respective ranges to find their values at which the objective functions listed in (4) are minimized. The variables contained within the objective calculations include power density P_{density} , mechanical power P_{mech} , machine length L , outer stator radius r_{so} , efficiency η , power loss P_{loss} , and torque ripple T_{rip} . Maximum torque T_{max} , minimum torque T_{min} and average torque T_{avg} , are the maximum, minimum, and average torque over a single electrical cycle.

$$\begin{aligned}
 O_1 &= -P_{\text{density}} = -\frac{P_{\text{mech}}}{V_{\text{machine}}} = -\frac{P_{\text{mech}}}{\pi r_{\text{so}}^2 L} \\
 O_2 &= -\eta = -\frac{P_{\text{mech}}}{P_{\text{mech}} + P_{\text{loss}}} \\
 O_3 &= T_{\text{rip}} = \frac{T_{\text{max}} - T_{\text{min}}}{T_{\text{avg}}}
 \end{aligned} \quad (4)$$

B. Design Evaluation

The process for each design evaluation step is summarized by the flowchart shown in Fig. 7. The numbered blocks in this diagram are now described in detail.

1) *Structural Analysis*: Each design within the population undergoes structural analyses to compute the maximum stress within the rotor-cross section using JMAG's 2D FEA structural

solver module at four different shaft speeds. These speeds are determined as 1/4, 1/2, 3/4, and 1 times a theoretical maximum speed for each rotor type. This theoretical maximum speed is taken as 20,000 RPM for the baseline rotor and 60,000 RPM for the MMAM and vision rotors, based on the yield strength of each material and maximum conventional tip speeds [4].

2) *Max. Speed Computation*: The maximum allowable rotational speed is found through a polynomial curve fit. The four operation speeds from the previous step constitute the x-values for the curve fit. Their associated maximum stress values constitute the y-values. The polynomial fitted to the data is found and the maximum speed is determined by interpolating the polynomial to find the speed where the 460 MPa yield stress from Table I is reached. A 1.5x safety factor is applied to determine the true maximum speed of each design.

3) *Electromagnetic Analysis*: For each design, the SynR machine is modeled and analyzed using JMAG's transient 2D FEA solver. The machine is evaluated at the maximum torque per ampere condition over a single electrical cycle/rotation. The "rated" rotor speed is set to the free variable ω_{frac} times the true maximum speed determined in Step 2. The outputs of this analysis include the shaft torque and losses.

4) *Evaluate Objectives*: The objectives (4) are calculated through a series of post-processing steps using the results from the electromagnetic analysis. The total losses P_{loss} include ohmic and iron losses combined with windage loss found via [29], calculated based on the rotor speed and air gap length. The post-processing steps take the raw torque and cumulative loss data and compute each objective for each rotor design.

VI. RESULTS AND DISCUSSION

This section presents results from the design space exploration of the baseline, MMAM, and vision SynR machine design variants using the multi-physics optimization framework developed in Section V to understand the general design trends between the three variants and to draw conclusions on the potential impact of MMAM on SynR machine design.

A. Optimization Results

The final pareto fronts obtained from the design space exploration study for the baseline, MMAM, and vision geometries are plotted in Fig. 8. The MMAM and vision designs are significantly more power-dense, while the efficiency of the conventional design remains higher at its maximum speed. This can be attributed to the relatively high rotational speeds of the MMAM and vision designs (approximately 3 - 4x that of the baseline design) enabled by the solid flux barriers. The higher rotational speeds improve output power and consequently power density for the MMAM and vision designs while also increasing the speed-dependent windage and iron losses. The lower efficiency and power density of the MMAM design compared to the vision design can be attributed to the non-ideal flux barrier shape (a limitation of available MMAM equipment). These results show that despite the non-ideal flux barrier shape, the MMAM designs still offer approximately 3x power density improvement compared to baseline designs.

TABLE II
OPTIMIZATION VARIABLES AND SELECTED DESIGNS

Parameter	Range	Baseline	MMAM	Vision	Units
α_b	[135, 150]	135	n/a	135	deg
d_{r1}	[3.2, 16]	3.692	4.799	3.845	mm
d_{r2}	[4.8, 16]	6.441	9.931	6.471	mm
l_{b1}	[20, 30]	29.98	n/a	29.97	mm
l_{b2}	[21.6, 32.4]	21.67	n/a	21.63	mm
l_{b4}	[9.6, 14.4]	13.60	n/a	14.28	mm
l_{b5}	[4.8, 7.2]	6.763	n/a	6.594	mm
r_{ri}	[3, 12]	11.46	6.348	11.31	mm
r_{ro}	[45, 50]	49.98	49.99	49.99	mm
w_{b1}	[2, 8]	4.763	5.543	4.783	mm
w_{b2}	[2, 8]	4.669	3.609	4.697	mm
ω_{frac}	[60, 100]	98.4	65.5	68.8	%

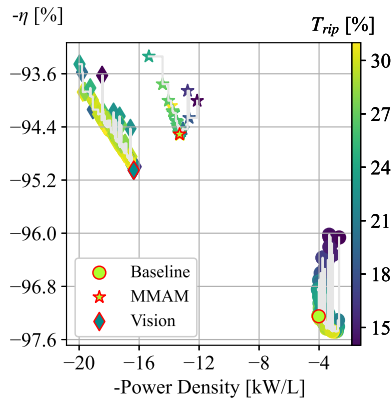


Fig. 8. Pareto fronts of SynR machine designs. Note that the objectives (4) are calculated at each design's rated conditions (speed and load).

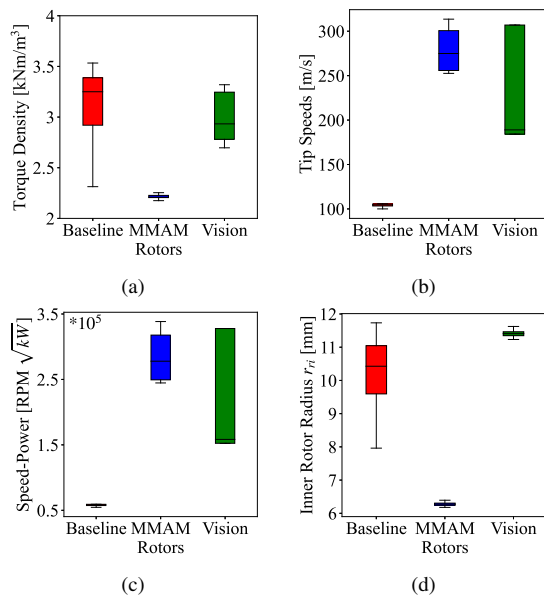


Fig. 9. Parameter distributions for: (a) tip speed, (b) speed-power metric, (c) torque density, and (d) r_i for rotor designs.

A maximum power density of approximately 16 kW/L can be seen for the MMAM rotor designs in Fig. 8, while the vision rotor designs shows values approaching 20 kW/L. This indicates a 3-4x power density improvement over the baseline design. The torque densities, seen in Fig. 9a, show a lower torque distribution from the MMAM rotor. This suggests that the current MMAM restrictions limit their maximum saliency ratios, reducing the torque that can be generated. A distribution of the tip speeds for each rotor design is shown in Fig. 9b. The optimal design for the baseline rotor results in a geometry that can only attain a maximum tip speed of 110 m/s. The MMAM and vision rotors are able to achieve much higher tip speeds, reaching up to 300 m/s. This suggests that despite the lower saliency ratios, the MMAM design can still achieve significantly higher power densities due to its increased speed capability. In Fig. 9c, analyses of the speed-

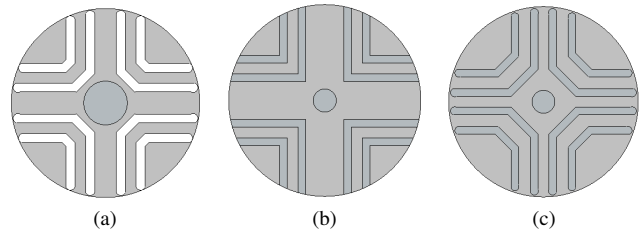


Fig. 10. Rotor cross-sections for: (a) baseline, (b) MMAM, and (c) vision rotor designs.

TABLE III
PERFORMANCE OF SELECTED DESIGNS

Parameter	Baseline	MMAM	Vision	Units
Power Density	4.0	13.3	16.4	kW/L
Torque Ripple	31.6	27.7	21.8	%
Efficiency	97.2	94.5	95.1	%
Maximum Speed	20,244	57,473	57,414	RPM
Tip Speed	106	301	301	m/s

power metric indicate a maximum speed-power metric of almost $3.5 \times 10^5 \text{ RPM}\sqrt{\text{kW}}$ for the MMAM and vision rotors, while the baseline rotor only achieved a maximum value of $0.6 \times 10^5 \text{ RPM}\sqrt{\text{kW}}$. This indicates the ability for the MMAM and vision rotors to generate greater power levels at faster rotational speeds. The MAMM design optimizes at a smaller inner rotor radius than the baseline and vision designs, seen in Fig. 9d. A larger rotor inner radius in the MMAM rotor would introduce interference between the rotor inner surface and the inner-most flux barrier, restricting any flux to pass through this region, which would decrease its efficiency and power density.

B. Selected Designs

To understand how the design space trends translate into actual machines, three promising designs are identified, one from each rotor variant. These designs are indicated with a red outline in Fig. 8, with the free variable values listed in Table II. These designs are selected based on trade-offs between the objectives. Designs with the highest power density were immediately prioritized since the hypothesis of this study is that power density can be improved with MMAM. However, designs with high power density but lower efficiencies were avoided, since sacrificing efficiency for power density would not be practical. The lowest torque ripple values were preferred, but did not affect the selection of designs.

Rotor cross-sections for the selected designs are shown in Fig. 10 and their performance parameters are listed in Table III. In Fig. 10, significant resemblance is seen between the baseline and vision rotors, indicating the flux barrier shapes for optimal torque production (and thus power density and efficiency) are similar, irrespective of flux barrier material.

The stress distributions for each selected rotor are plotted in Fig. 11 at their maximum operating speed (see Table III). Stress concentrations in the baseline design are significant between the flux barriers and outer edge of the rotor, limiting the maximum operating speed. The addition of the solid flux barrier significantly reduces the stress concentrations.

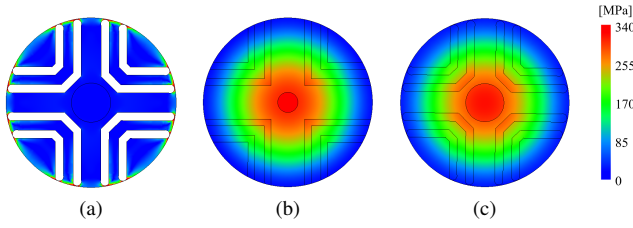


Fig. 11. Stress distributions for: (a) baseline, (b) MMAM, and (c) vision rotor designs at each design's rated speed.

TABLE IV
LOSS DISTRIBUTIONS ¹

Losses	Baseline	MMAM	Vision	Units
Stator Iron	196.2	254.5	200.6	W
Rotor Iron	37.81	9.510	35.35	W
Ohmic	57.19	57.19	57.19	W
Windage	128.5	128.5	128.5	W

¹ Calculated at 20,244 RPM.

Both the MMAM and vision rotor designs exhibit comparable maximum speeds since their geometry is simply the same solid metal structure but configured with different arrangements of Fe-3.5Si and 316L.

The loss mechanisms for each rotor are seen in Fig. 12a and Table IV. Since several loss components are speed dependent, each selected design is re-analyzed at the same speed (baseline design speed) to facilitate accurate comparisons. Notable is the difference in the iron losses between the MMAM and baseline/vision designs. The stator iron losses appear higher in the MMAM but rotor iron losses appear lower. This is assumed to be due to the alignment between the stator teeth and rotor flux barriers based on the restrictions of the MMAM rotor. However, this did increase the overall iron losses in the MMAM design. The windage and ohmic losses appear consistent, which is to be expected. As their speed increases beyond the baseline rotor's limit, the MMAM and vision rotors see an increase in stator iron and windage losses.

From Table III, the MMAM and vision rotors have the ability to achieve significantly higher power density, due to their ability to run at higher speeds than the baseline rotor. The tip speed improvement from the baseline rotor to the MMAM and vision rotors are approximately 3x. This leads to a 3-4x improvement in power density for the MMAM and vision rotors, despite existing MMAM geometry restrictions.

Machine efficiencies (O_2 in (4)) are calculated using the resulting power and losses from each design state to compare rotor performances up to each design's maximum speed. The resulting efficiency profiles are shown in Fig. 12b. The baseline and vision rotors exhibit similar efficiency curves, which is expected as their cross-sections are very similar, with the main difference being the flux barrier material. The vision rotor can run much faster where it only then becomes less efficient. The MMAM rotor exhibits a slightly lower efficiency profile, which is expected as the restrictions in the MMAM design geometry have limited its saliency ratio, which limits its torque production (confirmed in Fig. 9a and seen in Table V).

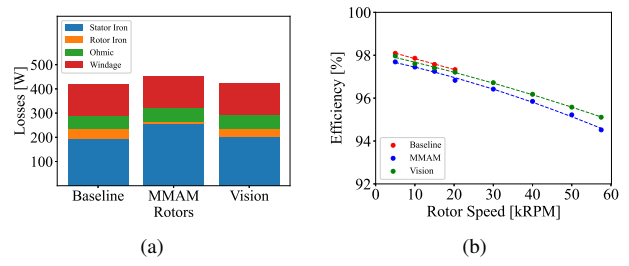


Fig. 12. Performance parameters for selected designs: (a) loss components at 20,244 RPM and (b) efficiency profile.

Finally, Table V compares the three selected SynR designs in this study with conventional electric machine types. Data for the conventional machines in Table V was found from [7], [11], [30], [31], and through calculations per (3). It should be noted that the MMAM and vision rotors exceed the maximum tip speed of all other machines through their solid-cylindrical cross-section. The implementation of MMAM in SynR rotors allows them to exceed the highest power densities in conventional SynR, surface PM, and induction machines, confirming the theory proposed in Section III-A that SynR power density can be increased with MMAM.

VII. CONCLUSIONS

This paper developed and proposed one of the first SynR rotor designs that uses new MMAM concepts to improve SynR machine power density. The investigation involves i) the fabrication and characterization of single- and multi-material test specimens using LPBF, ii) the design of rotor structures with and without existing geometric restrictions, and iii) the simulation and optimization of three different SynR rotor geometries using the characterized sample data. The core innovation lies in the substitution of air in rotor flux barriers with solid, non-magnetic steel, which leads to enhanced rotor strength, facilitating the realization of higher tip speeds and increased power densities.

The experimental process involves the fabrication of LPBF test samples using Fe-3.5Si and 316L stainless steel, thereby facilitating an assessment of relevant properties and multi-material bond strength. Results indicate the materials can bond successfully, making an interfacial bond stronger than the yield strength of the weaker material. Magnetic properties of additively manufactured Fe-3.5Si were also measured. The results reveal a BH-curve comparable to M19 29 Gauge electric steel, affirming the feasibility of employing additively manufactured Fe-3.5Si.

Rotor designs with (MMAM design) and without (vision design) existing LPBF restrictions were put forward and optimized to increase power density and efficiency. This paper finds that the MMAM and vision rotor geometries can achieve efficiencies of $\geq 94.5\%$ at tip speeds of > 300 m/s, which indicates a 300 % increase in tip speed over a similarly-optimized baseline conventional rotor design. All designs exhibited similar efficiencies at the speeds which the baseline design could reach. The results from the design space

TABLE V
COMPARISON OF SYN R ROTORS TO CONVENTIONAL ELECTRIC MACHINES

Parameter	Baseline	MMAM	Vision	Conventional SynR	Surface PM	Induction	Units
Power Density	4.0	13.3	16.4	≤ 10	≤ 10	≤ 10	kW/L
Speed-Power Metric	6.9×10^4	3.4×10^5	3.2×10^5	$\leq 3.4 \times 10^5$	$\leq 8 \times 10^5$	$\leq 6 \times 10^5$	RPM $\sqrt{\text{kW}}$
Saliency Ratio	5.91	4.48	5.95	≤ 8	1	1	-
Maximum Tip Speed	106	301	301	130-170	200-250	150-200	m/s
Air Gap Shear Stress	17.6	19.9	24.8	≤ 33.5	≤ 36.0	≤ 21.5	kPa

exploration and optimization showed that MMAM technology is capable of improving the tip speed of SynR rotor designs by up to 300 % to affect power density improvements of over 400 %, both of which are very significant improvements despite the restrictions that currently exist with the MMAM process. Ultimately, the results in this paper demonstrate the potential to fabricate SynR rotors with solid metallic flux barriers using MMAM and position SynR machines with solid rotors as compelling candidates to realize high-speed and power-dense SynR designs. These results motivate future work to fabricate a prototype and experimentally validate its performance.

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